

Kai S. Yun

Legal Name: SirkHoo Yun

Ph.D. Student, Laboratory for Information & Decision Systems (LIDS) at MIT
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RESEARCH INTEREST

Leveraging the intersection of **control theory**, **deep learning**, and **foundation models** to develop and deploy **safe, scalable controllers** on **physical robots**, such as humanoids, quadrupeds, and drones.

EDUCATION

Massachusetts Institute of Technology
Ph.D. in Mechanical Engineering

Class of 2029 (Expected)
Cambridge, MA

Carnegie Mellon University
M.S. in Mechanical Engineering

Class of 2025
Pittsburgh, PA

University of California, Berkeley
B.S. in Mechanical Engineering, Minor in EECS

Class of 2023
Berkeley, CA

HONORS & AWARDS

MathWorks Fellowship, 2026 — 2027, MIT School of Engineering
ATK-Nick G. Vlahakis Fellowship, 2024 — 2025, CMU College of Engineering
Best Master's Research Paper Commencement Award, 2025, CMU Department of Mechanical Engineering
Conference Grants: MIT OGE (2026), CMU OGPA (2024), CMU ME (2024)

PUBLICATIONS

P: preprints, C: conference proceedings, J: journal articles, *equal contribution

- [P1] **Kai S. Yun**, Zeyang Li, Navid Azizan, “Provably Safe, Yet Scalable Reinforcement Learning.” Preprint at [arxiv.org:2606.14536](#), 2026.
- [C2] **Kai S. Yun**, Navid Azizan, “ATOM-CBF: Adaptive Safe Perception-Based Control under Out-of-Distribution Measurements.” Published in *Learning for Dynamics and Control Conference (L4DC)*, 2026. Poster Presentation in Los Angeles, California, 2026. Full paper at [arxiv.org:2511.08741](#).
- [C3] Yifan Sun, Rui Chen, **Kai S. Yun**, Yikuan Fang, Sebin Jung, Feihan Li, Bowei Li, Weiye Zhao, Changliu Liu, “SPARK: Safe Protective and Assistive Robot Kit.” Published in *IFAC Symposium on Robotics*, 2025. Full paper at [arxiv.org:2502.03132](#), 2025, [Website](#).
- [C4] **Kai S. Yun**, Rui Chen, Chase Dunaway, John M. Dolan, Changliu Liu, “Safe Control of Quadruped in Varying Dynamics via Safety Index Adaptation.” Published in IEEE *International Conference on Robotics and Automation (ICRA)*, 2025. Oral Presentation in Atlanta, Georgia, 2025, [Website](#).
- [C5] Simin Liu*, **Kai S. Yun***, John M. Dolan, Changliu Liu, “Synthesis and Verification of Robust-Adaptive Safe Controllers.” Published in *European Control Conference (ECC)*, 2024. Oral Presentation in Stockholm, Sweden, 2024.
- [C6] Tianhao Wei, Hanjiang Hu*, Luca Marzari*, **Kai S. Yun***, Peizhi Niu*, Xusheng Luo, Changliu Liu, “ModelVerification.jl: a Comprehensive Toolbox for Formally Verifying Deep Neural Networks.” Published in *International Conference on Computer Aided Verification (CAV)*, 2025, [GitHub](#).
- [J7] Jack C. Harms, Ethan M. Grame, **SirkHoo Yun**, Bushra Ahmed, Leah C. O'Brien, James J.

O'Brien, "Mass-independent Dunham Analysis of the $[7.7]Y^2\Sigma^+ - X^2\Pi_i$ and $[16.3]A^2\Sigma^- - X^2\Pi_i$ Transitions of Copper Monoxide, CuO." Published in *Journal of Molecular Spectroscopy*, 2019.

RESEARCH EXPERIENCE

Laboratory for Information & Decision Systems (MIT)

Aug 2025 — Present

Advisor: Prof. Navid Azizan

Cambridge, MA

- Vision-language-action (VLA) models for safe, scalable, and general-purpose robotics.
- Hard-constrained safe reinforcement learning for high-dimensional robots [P1].
- Perception-based safe control of robots for out-of-distribution measurements [C2].

Robotics Institute (CMU)

Jun 2023 — Jul 2025

Advisors: Prof. Changliu Liu, Prof. John M. Dolan

Pittsburgh, PA

- Safe humanoid autonomy and teleoperation toolbox (*SPARK*) (Unitree G1 platform) [C3].
- Adaptive safe controller for quadrupeds in varying dynamics (Unitree Go2 platform) [C4].
- Algorithm for synthesizing robust-adaptive safe controllers under uncertainty [C5].
- *ModelVerification.jl*, a Julia library containing state-of-the-art DNN verification algorithms [C6].

Berkeley AI Research Lab (UC Berkeley)

Aug 2021 — Apr 2022

Advisor: Prof. Koushil Sreenath

Berkeley, CA

- *PPO-Worst-Case*, a novel safe reinforcement learning algorithm for worst-case scenarios.
- Simulations for legged robots in safety-critical scenarios.

PROFESSIONAL EXPERIENCE

Tesla, Inc.

May 2022 — Aug 2022

Vehicle Dynamics / Software Engineering Intern

Palo Alto, CA

- Developed a framework for comparing subjective vehicle evaluation and objective test data.
- Tested and analyzed vehicle dynamics performance of Models S, 3, and Semitruck.
- Developed automated internal ticketing, reporting, and logging systems.

NeuroCore.ai

Oct 2020 — Jul 2021

Reinforcement Learning Engineering Intern

Seoul, South Korea

- Designed and developed RL training and deployment frameworks; 84% increase in training efficiency.
- Developed simulators for Supply Chain Management (SCM) tasks for semiconductor manufacturers.
- These are currently deployed in South Korean semiconductor manufacturers, such as SK Hynix.

TEACHING & SERVICE

Robotics Institute at CMU

Pittsburgh, PA

Manager/Developer — [RoboticsKnowledgebase](#) (Funded by Google)

Oct 2024 — Jul 2025

Teaching Assistant — F1Tenth Autonomous Racing (16.663)

Jan 2025 — May 2025

Robot Manager, Lab Manager — [Intelligent Control Lab](#)

Jan 2025 — May 2025

Lab Server Manager — [DRIVE Lab](#)

Jun 2024 — May 2025

Research Mentor — [Robotics Institute Summer Scholars \(RISS\)](#)

May 2024 — Aug 2024

Mentee: Chase Dunaway (Undergraduate at New Mexico Tech → Ph.D. at MIT)

Mechanical Engineering Department at CMU

Pittsburgh, PA

Teaching Assistant — Dynamic Systems and Controls (24.352)

Aug 2024 — Dec 2024

Mechanical Engineering Department at UC Berkeley

Teaching Assistant — Statistics and Data Science for Engineers (ENG 178)

Berkeley, CA

Jan 2023 — May 2023

Republic of Korea Army

Squad Leader, Sergeant, K-1 Tank Mechanic — Combat Service Support (CSS)

South Korea

Jan 2019 — Aug 2020

Reviewer: IEEE Robotics and Automation Letters**SKILLS**

Expertise: Humanoid, Quadruped, Drone, ML/RL, VLA, World Model, Safe Control, Optimization**Physical Robots/Machines:** Unitree G1, Go2, Pixhawk Drones, CrazyFlie, Sawyer, F1Tenth, Tanks**Libraries:** JAX, PyTorch, MJX, Gazebo, Gym, Isaac, PyBullet, Mosek, CasADi, PX4, Numpy**Programming/OS/Tools:** Python, MATLAB, Julia, C++, ROS, ROS2, Linux, Git, Docker**Other skills:** Jetson, OptiTrack, Vicon, OpenSim, SolidWorks, L^AT_EX, Simulink, Scuba Diving**Languages:** Fluent in English & Korean**SELECTED COURSES**

Carnegie Mellon UniversityProvably Safe Robotics (*16.883*), Optimal Control & Reinforcement Learning (*16.745*), Advanced Control Systems Integration (*24.774*), Engineering Optimization (*24.785*), Biomechanics of Human Movement (*24.663*).**UC Berkeley**Deep Reinforcement Learning (*CS 285*), Nonlinear Systems (*EE C222*), Machine Learning (*CS 189*), Robotic Manipulation & Interaction (*EECS C106B*), Vehicle Dynamics and Control (*ME 131*), Dynamic Systems & Feedback (*ME 132*), Mechatronics Design (*ME 102B*), Experimentation and Measurements (*ME 103*).**WORK AUTHORIZATION**

US Permanent Resident (Green Card holder), South Korean Citizen (military service completed)